KFUPM CAMPUS SHADE NAVIGATION SYSTEM

CoolPath

Finding the coolest, most shaded path — not just the shortest

Prepared for: Osamah | April 2026 | KFUPM

THE PROBLEM

Extreme summer heat makes campus walks a health hazard. No existing tool routes for shade.

45°C+

Peak summer temperature on KFUPM campus

5,088m

Exposed walkway distance across campus network

Heat Stress

Students exposed to dangerous heat during peak hours (10am–3pm) with no routing guidance.

No Accessibility Routing

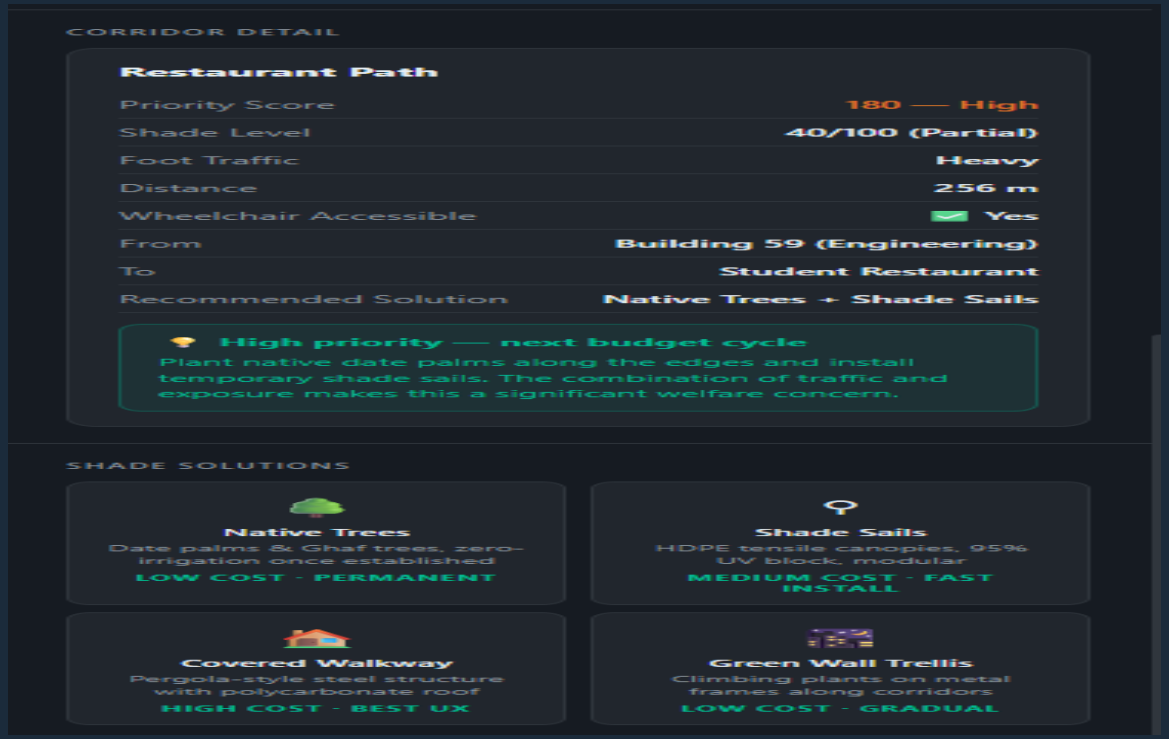
No path finder considers ramps, shade, rest stops, or wheelchair-accessible corridors.

No Planning Data

Facilities teams lack a heat-exposure priority map to guide shade infrastructure investment.

WHAT IS COOLPATH?

A Browser-Based Shade Navigation App



Shortest Route

Dijkstra's algorithm for fastest path

Coolest Route

Shade-weighted algorithm, comfort first

Accessible Route

Excludes inaccessible paths, step-free

Shade Plan Page

Infrastructure priority dashboard

THE ALGORITHM

How the Coolest Route is Calculated

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Coolest Weight = distance × ( 1 + 2.5 × (100 – shadeScore) / 100 )
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Fully Shaded

shadeScore = 100

× 1.0 effective cost

Partially Shaded

shadeScore = 60

× 2.0 effective cost

Fully Exposed

shadeScore = 0

× 3.5 effective cost

Coolest Route

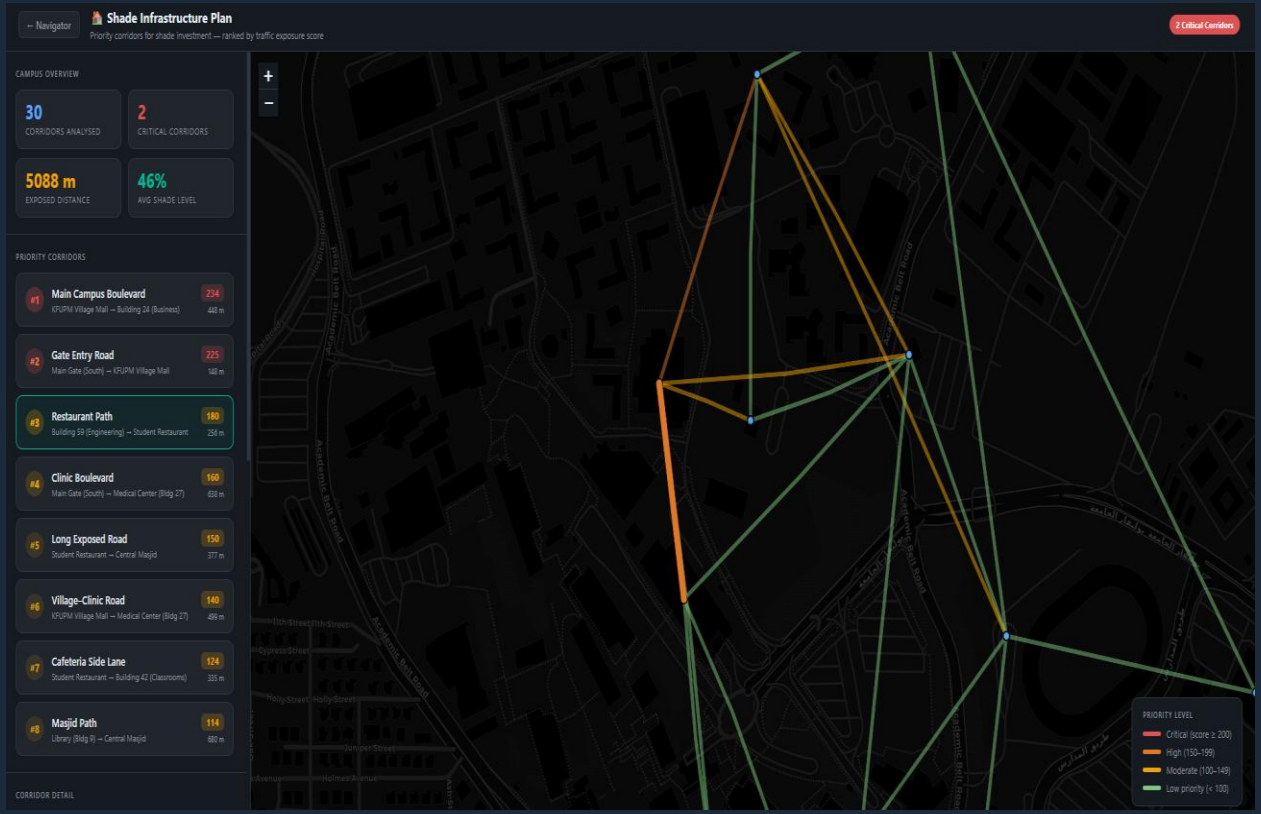
Prefers shaded corridors even if slightly longer in metres. Minimises heat exposure. Best for 10am–3pm peak hours.

Shortest Route

Ignores shade entirely. Minimises physical distance only. May route through fully exposed corridors at dangerous times.

LIVE DEMO

Navigator in Action: Building 76 to Student Restaurant



Route Card Stats

715m total · 56% shade
9 min walk time · comfort score

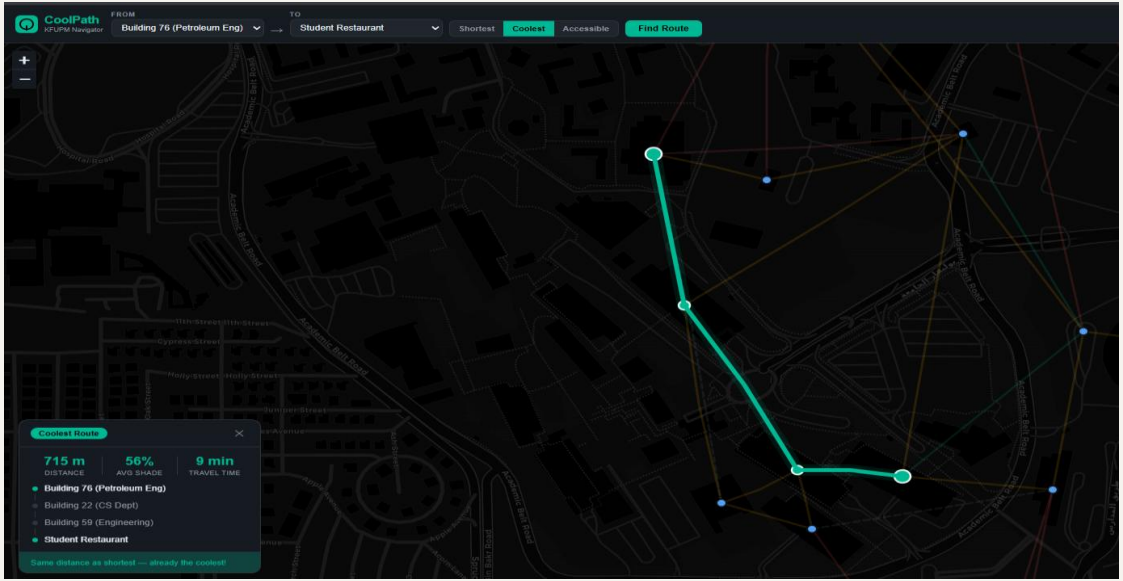
3 Map Tile Modes

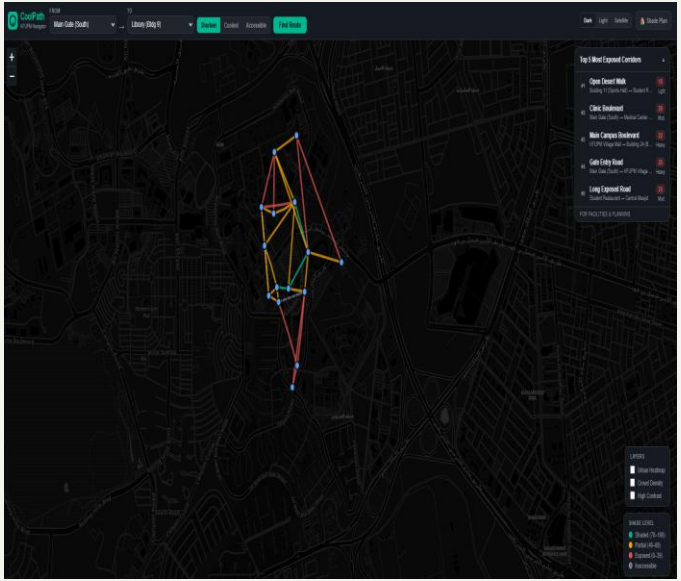
Street · Satellite · Campus
Switch live during routing

Overlay Layers

Heatmap · Crowd density
Toggle shade corridors

Infrastructure Priority Planning for Facilities Teams





$$\text{Priority Score} = \text{Traffic Density} \times (100 - \text{Shade Score})$$

- >= 200 — Critical Priority
- 150–199 — High Priority
- < 100 — Low Priority

INSTITUTIONAL BACKING

Vision 2030 Alignment

Disability Inclusion

- CRPD obligation — Saudi Arabia ratified 2008
- National Transformation Program KPI: inclusive infrastructure
- KFUPM Policy obligates accessible routing
- Accessible mode already built into CoolPath

Liveability & Wellbeing

- Quality of Life Program targets outdoor comfort
- Shade = direct heat-health intervention
- Supports walkability goals in hot climate
- Reduces reliance on air-conditioned transport

Sustainable Urban Design

- Native tree corridors reduce urban heat island
- Data-driven shade infrastructure investment
- Replicable model for other Saudi campuses
- Aligns with NEOM & Vision 2030 city design

ACCESSIBILITY

Disability-Inclusive Routing

How Accessible Mode Works

- 1. Graph rebuild — inaccessible edges removed from routing graph
- 2. Shade weighting still applied — comfort remains a priority
- 3. Steps, steep slopes, unpaved surfaces excluded automatically
- 4. Rest-point nodes flagged for future waypoint integration
- 5. High-contrast mode + screen-reader labels already implemented

Who Benefits Most

Wheelchair users

Avoids steps, rough surfaces, inaccessible gates

Elderly

Shorter exposure time + rest points on route

Chronic conditions

Heart/respiratory — heat exposure is life risk

Low-vision users

High-contrast mode + full screen-reader support

Deaf users

Visual-only alerts, no audio dependency

CURRENT DATA

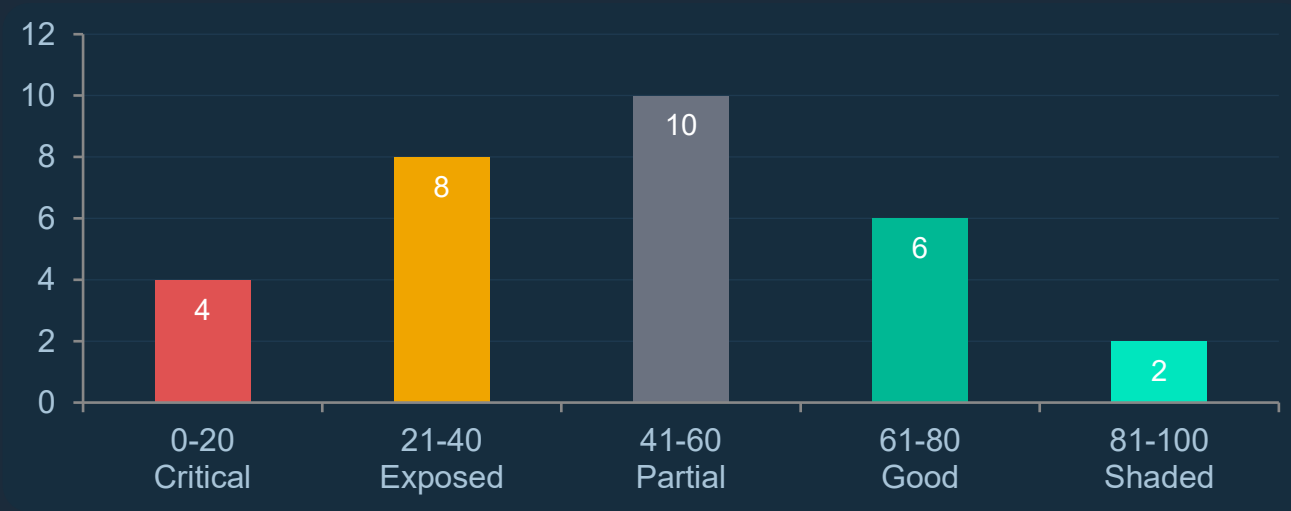
Campus Network at a Glance

15
Buildings GPS-verified

30
Corridors Mapped

46%
Average Shade Score

5,088m
Exposed Walkway



Key Insight

12 of 30 corridors score below 40 — 40% of the network rated Exposed or worse.

COLLABORATION OPPORTUNITY

ARE Department: Four Data Upgrades

Solar Shadow Simulation

Ladybug + Grasshopper tools generate hour-by-hour shade maps.
Replaces manual estimates with high accuracy.

Impact: From ~70% to ~95% shade accuracy

UTCI Thermal Comfort

Universal Thermal Climate Index — gold standard outdoor comfort metric.
Enables academic publication and policy citation.

Impact: Publishable methodology

Surface Material Survey

Pavement type (concrete vs asphalt vs tile) affects radiant heat.
Different emissivity values refine routing weights.

Impact: More precise coolest-route calculation

Live Microclimate Sensors

Real-time temperature + humidity feeds from campus weather stations.
Enables dynamic routing that updates through the day.

Impact: Real-time comfort routing

NEXT STEPS

Three-Phase Development Roadmap

01

This Semester

- ARE solar shadow data integration
- Expand to 25+ buildings campus-wide
- Time-of-day shade variation
- Waypoint GPS accuracy improvements

02

Next Semester

- UTCI-based route weighting
- Rest point nodes on accessible routes
- Elevator + ramp link integration
- Mobile-responsive layout optimisation

03

Research & Publication

- Live microclimate sensor feeds
- Multi-university campus model
- ML-based shade score prediction
- Academic paper submission

THREE THINGS TO REMEMBER

CoolPath: Ready. Waiting for Real Data.

1 Shortest path is not safest path at 45°C. CoolPath solves a real welfare problem no existing tool addresses.

2 Vision 2030's disability inclusion and sustainability mandates give this institutional backing — not just academic interest.

3 With ARE department data, this becomes scientifically rigorous and publishable. The tool is ready — it needs your data.